

Education

University College London

Doctor of Philosophy, 2026.

PhD thesis: Layered Monoidal Theories.

Research group: Programming Principles, Logic and Verification.

Supervisor: Fabio Zanasi.

University of Amsterdam

MSc Logic (cum laude), 2020.

MSc Thesis: Quantum quirks, classical contexts: Towards a Bohrification of effect algebras.

Supervisors: Chris Heunen (University of Edinburgh), Nick Bezhanishvili.

University of Edinburgh

BSc Mathematics (First class honours), 2018.

BSc Project: Sheaves on Topological Spaces and their Logic.

Supervisor: Tom Leinster.

Jyväskylä Normaalikoulu (Finland)

Ylioppilastutkinto (High school matriculation examination), 2014.

Mathematics, Physics, Chemistry, Finnish, Russian, English, French.

Publications

A categorical model for organic chemistry

with Ella Gale, Fabio Zanasi.

Theoretical Computer Science 1032 (2025).

doi: 10.1016/j.tcs.2025.115084.

Learning closed signal flow graphs

with Ekaterina Piotrovskaya, Fabio Zanasi.

Theoretical Aspects of Computing – ICTAC (2024).

doi: 10.1007/978-3-031-77019-7_5.

Disconnection rules are complete for chemical reactions

with Ella Gale, Fabio Zanasi.

Theoretical Aspects of Computing – ICTAC (2024).

doi: 10.1007/978-3-031-77019-7_13.

A categorical approach to synthetic chemistry

with Ella Gale, Fabio Zanasi.

Theoretical Aspects of Computing – ICTAC (2023).

doi: 10.1007/978-3-031-47963-2_17.

String diagrams for layered explanations

with Fabio Zanasi.

Electronic Proceedings in Theoretical Computer Science, 380 (2023).

doi: 10.4204/EPTCS.380.21.

There and back again: A circuit extraction tale

with Miriam Backens, Hector Miller-Bakewell, Giovanni de Felice, John van de Wetering.

Quantum 5, 421 (2021). doi: 10.22331/q-2021-03-25-421.

Statistical Hypothesis Testing in the Context of Hume’s Critique of Induction

University of Edinburgh Philosophy Society Journal, Issue 2: 2017-18.

Conference talks

International Colloquium on Theoretical Aspects of Computing 2024

Disconnection rules are complete for chemical reactions.

Applied Category Theory 2024

Disconnection rules are complete for chemical reactions.

Talk recording: youtu.be/JdeDkUE92Uo?.

International Colloquium on Theoretical Aspects of Computing 2023

A categorical approach to synthetic chemistry.

Tenth Symposium on Compositional Structures 2022

Quantum quirks, classical contexts: Towards a Bohrification of effect algebras.

Slides: cl.cam.ac.uk/events/syco/10/slides/lobski.pdf.

Applied Category Theory 2022

String diagrams for layered explanations.

Talk recording: youtu.be/iNdH_U-C1Nc?t=22803.

Quantum Physics and Logic 2020

There and back again: A circuit extraction tale.

Talk recording: youtu.be/Orilw6ujWag.

Refereeing

Journal referee

- Compositionality. compositionality.episciences.org.
- Quantum. quantum-journal.org.

Conference referee (external)

- European Symposium on Programming (ESOP). 2026.
- Theory of Quantum Computation, Communication and Cryptography (TQC). 2023.
- International Colloquium on Automata, Languages, and Programming (ICALP). 2023.
- Applied Category Theory (ACT). 2022.
- Quantum Physics and Logic (QPL). 2021.

Projects and Work experience

University College London

Research Fellow.

October 2025 - Present.

University College London

Various teaching assistant roles in mathematics and theoretical computer science.

January 2022 - December 2025.

Aboa Vetus Ars Nova museum

Civilian service year.

October 2020 - September 2021.

University of Amsterdam

Teaching assistant for MSc Mathematics course “Category Theory and Topos Theory”.
Autumn 2019.

Tartu Observatory

Development of Satellite Attitude Determination Software.

Summer 2018.

London Mathematical Society Undergraduate Research Bursary

Project title: Quantum Logic and Set Theory.

Report title: The Category of Hilbert Spaces as an Orthogonal Category.

Project blog: andinparticularexists.wordpress.com.

Supervisor: Chris Heunen.

Summer 2017.

Tartu Observatory

Erasmus+ traineeship.

Traineeship title: Development of Satellite Attitude Determination Software.

Summer 2016.

Department of Physics, University of Jyväskylä

Summer trainee, 2013.

Extracurricular
positions

The University of Edinburgh SocieTea

Committee position, Publicity, 2017-18.

Committee position, President, 2016-17.

The University of Edinburgh Philosophy Society

Committee position, Ordinary member, 2015-16.

Other
experience

Fibred category theory and type theory reading group

Organizer, 2022-23.

Infinity category theory reading group

Organizer, 2019.

Reading project in point-free topology

Supervisor: Nick Bezhanishvili.

2019-2020.

Research project in categorical models for homotopy type theory

Supervisor: Benno van den Berg.

2019-2020.

Amsterdam Fringe Festival

Volunteer, September 2019.

Volunteer, September 2018.

Logic and Physics reading group

Organizer, 2018-2019.

Hidden Door Festival

Volunteer, May-June 2018.

Art-Master Theatre

Website translator, occasionally between 2013 and 2015.

Languages

Finnish (native), Russian (native), English (fluent),
French (DELFI B2), Swedish (intermediate).